

CHRISTOPHER AL-RAHI

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EDUCATION

University of Ottawa

BASc. in Mechanical Engineering | BSc. in Computing Technology (Co-op)

Ottawa, ON

September 2022 - April 2027

EXPERIENCE

Curtiss-Wright Defense

Mechanical Engineering Intern

Ottawa, ON

September 2025 - December 2025

- Independently executed thermal redesign of liquid-cooled LFT modules using Simcenter FloTHERM, reworking internal coolant channel geometry to improve heat dissipation, and presented simulation results and design recommendations.
- Assembled and validated eight ruggedized electronic chassis for defense systems, ensuring compliance with military-grade reliability and quality standards.
- Performed environmental and thermal reliability testing on nine conduction-cooled products (4000+ insertions, 80+ thermal cycles) and reported product qualification and design verification.

CanmetENERGY - Alternative Energies Laboratory

Research Assistant

Ottawa, ON

January 2025 - April 2025

- Designed four ventilation and mechanical CAD layouts across three projects, reducing pressure loss and optimizing system efficiency.
- Automated heat pump performance analysis with VBA, accelerating the process by 90%.
- Heat and pressure loss calculations performed to refine mechanical design and improve system reliability.

CanmetENERGY - Renewable Heat & Power

Research Assistant

Ottawa, ON

May 2025 - August 2025

- Conducted root-cause analysis and restored the motion of a solar actuator system from $\pm 15^\circ$ to $\pm 40^\circ$, improving the tracking control and reliability of the system through mechanical redesign.
- Designed and implemented a height-adjustable shielded sensor system, increasing live temperature measurement accuracy by over 30%.
- Designed and built a compact, portable ventilation prototype for solar thermal testing, optimizing airflow and integration within tight spatial constraints.

PROJECTS

ArcAI - Mobile Shooting Robot (In Progress) *Computer Vision, Machine Learning, Embedded Control*

Building a tracked autonomous robot integrating computer vision, laser sensing, and closed-loop control to detect and engage a visual target from arbitrary starting positions. Designed a mechanically repeatable launch subsystem optimized through analytical modeling and iterative prototyping.

Hot Tub Heater Element Repair *Failure Analysis*

Performed a structured fault analysis on a malfunctioning hot tub heating system, isolating electrical failures through diagnostic testing and circuit inspection. Researched and sourced compatible components, then removed and reinstalled the heater element within the piping assembly to restore full functionality.

Beer Goggles - Specific Gravity Measuring Device *Arduino, Python, SolidWorks*

Led a 3-person team to design and prototype a sensor-based measurement system (± 0.1 accuracy) with ultrasonic and load-cell integration; completed 3 iterative design cycles for calibration and validation.

AWARDS

Dean's Honour List | Recognition of Excellence

Awarded for maintaining a grade point average of 8.5 or better.

Faculty of Engineering

September 2022 | September 2024

SKILLS

Mechanical & CAD:	SolidWorks, AutoCAD, ANSYS, 3D Printing, FEA, Creo, GD&T
Embedded systems & Hardware:	Arduino, Raspberry Pi, Sensors & Actuators, DAQ, IMU Integration
Thermal & Energy Systems:	PID Control, HVAC Design, Pressure & Heat Load Calculations, Flotherm
Testing & Analysis:	Reliability Testing, Vibration Management, Environmental Qualification
Programming & Automation:	Python, VBA, C++, Arduino, Computer Vision (OpenCV), ML Inference
Robotics & Control Systems:	Autonomous Systems, Sensor Fusion, Motor Control, Mechatronics